

Alexander Kai Chen

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Mission

Exploring the intersection of computer science and classics to understand how language encodes thought. Building neurosymbolic and explainable systems informed by classical languages and philosophy to make reasoning transparent.

Publications

2025 — iOS as Acceleration

NeurIPS 2025 ER Workshop

- Paper (<https://arxiv.org/abs/2512.22180>) accepted as Poster into NeurIPS 2025 Efficient Reasoning Workshop, mentored under Roger Jin (prev. @ Nous Research, Apple, Google Student Researcher)
- Built novel distributed pipeline parallel framework from scratch for utilizing mobile phones as heterogeneous compute accelerators compatible with local consumer hardware

Skills

- **Programming Languages:** Python, Java, Swift, C++, Lisp (Clojure), Prolog, Lean4
- **Frameworks & Tools:** AWS, Azure, Angular, FastAPI, PyTorch, Mathlib
- **Workflows:** SCRUM, Agile
- **AI/ML:** Neural-Symbolic Systems, Explainable AI, Distributed Computing & Parallelism, Edge & Mobile Computing
- **Theory:** Discrete Mathematics, Formal Verification, Theorem Proving, Data Structures & Algorithms

Experience

2025 — SWE ET Internship

State Farm

- Summer 2025: Software Engineer Intern @ State Farm, Enterprise Technology division; AWS, Azure, Angular; SCRUM + Agile workflows

2024 — XAI Education Research Internship

UTD ALPS

- Summer 2024: Explainable AI educational research intern @ UT Dallas Applied Logic & Programming Systems lab, under Professor Gopal Gupta
- Explored modeling scenarios using Prolog, combining neural networks and symbolic logic systems (NeSy), implemented logic problem solvers

Honors & Achievements

2024-2025 — USACO Gold

USA Computing Olympiad, Competitive Algorithmic Programming

- Placed 1,022/12,591 world-wide in first contest in USA Computing Olympiad; advanced to Silver division in first year, Gold division following year

2025-2026 — ISEF State Fair

Texas Science & Engineering Fair

- Placed second in "Software Design / Systems Software" category, earning full scholarship to Governor's Science & Technology Champion's Academy summer research program

2025-2026 — ISEF Regional Fair

Dallas Regional Science & Engineering Fair

- Placed first, winning "Software Design / Systems Software" category with project "MobilePipe: iPhones as Parallel Compute Accelerators for Local Machine Learning" and advancing to State Fair.

2022-2025 — Speech & Debate

NCFCA Speech & Debate, Coolidge Foundation Debate

- Placed 2nd team policy speaker nation-wide at NCFCA online national open tournament
- Ranked as 2nd individual speaker at Coolidge Foundation national debate tournament

2022-2024 — FRC Robotics World Championships

FIRST Robotics Competition

- Served on robotics programming team & led scouting data analytics application development, advancing to world final championship for two years

Education

2023–Present — Highschool

Homeschool

- 4.0 GPA, 1600 SAT, 6x APs score 5, currently completing Junior year

2025–Present — Dual Credit College Courses

Collin College

- Discrete Math (Grade: A, 100%)
- Linear Algebra (in-progress)

2023–Present — Online College Courses

Online

- "Data Structures & Algorithms Specialization" @ UCSD
- "Build a Modern Computer from First Principles" @ HUJI